

O³Afford: One-Shot 3D Object-to-Object Affordance Grounding for Generalizable Robotic Manipulation

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Abstract—Grounding object affordance is fundamental to robotic manipulation as it establishes the critical link between perception and action among interacting objects. However, prior works predominantly focus on predicting single-object affordance, overlooking the fact that most real-world interactions involve relationships/contacts between pairs of objects. In this work, we address the challenge of object-to-object affordance grounding under limited data. Inspired by recent advances in few-shot learning with 2D vision foundation models, we propose a novel one-shot 3D object-to-object affordance learning approach for robotic manipulation. Semantic features from vision foundation models combined with point cloud representation for geometric understanding enable our one-shot learning pipeline to generalize effectively to novel objects and categories. We further integrate our 3D affordance representation with large language models (LLMs) to serve as planners for low-level actions, significantly enhancing LLMs’ capability to comprehend and reason about object interactions when generating task-specific constraint functions. Our experiments on 3D object-to-object affordance grounding and robotic manipulation demonstrate that our method significantly outperforms existing baselines in terms of both accuracy and generalization.

I. INTRODUCTION

Affordance grounding aims to identify the functional properties of objects or environments that indicate potential interactions, effectively communicating how objects can be manipulated and interacted with in meaningful ways. Accurate prediction of affordance maps for objects can significantly enhance numerous downstream tasks like robotic manipulation [1].

Although prior works [2, 3] have explored 3D object affordance grounding using object point clouds, they predominantly focus on single-object affordance prediction with the assumption that the affordance pertains solely to human interaction. This approach neglects the reality that many daily tasks involve object-to-object interactions, which represent more contact-rich challenges for robotic manipulation. For instance, the cutting task typically requires two objects: a source object (knife) and a target object (apple). Such an assumption limits the potential generalization ability to broader scenarios. We aim to study object-to-object affordance grounding in this work. A challenge in object-to-object affordance grounding is the limited annotated data for model training. O2O-Afford [4] addressed this problem in an annotation-free manner through automatically extracting contact in simulation, which renders it only applicable to simple affordances such as placing and fitting. Our work aims to infer more complex affordances such as pouring and cutting, which are unattainable for O2O-Afford yet crucial

in common robotic manipulation tasks.

We seek to develop a solution for generalizable object-to-object affordance grounding with minimal supervision. Recent advances in vision foundation models [5–7] have demonstrated impressive zero-shot generalization capabilities across various vision tasks. Being pre-trained on internet-scale datasets, these vision foundation models (VFMs) have been equipped with generalizable semantic understanding [8] and part awareness [9]. A series of works have explored leveraging the capabilities of VFMs for few-shot downstream vision tasks such as segmentation [10], detection [11], and visual correspondence [12]. Inspired by a prior work [13] in VFMs for 2D affordance, we further investigate VFMs for 3D affordance with the hypothesis that point clouds provide richer geometric information, thus enabling generalizable capabilities across different viewpoints, unseen object instances, and even entirely novel object categories.

In this paper, we introduce O³Afford, a One-shot Object-to-Object Affordance Grounding framework with 3D feature field distilled from VFMs. We first project features from DINOv2 [5] from multi-view observations onto the point clouds of both the source object and the target object to be manipulated. The two object point clouds, enriched with part-aware semantic features, are then processed through our bi-directional affordance discovery module, which considers the geometric contexts of both objects in both directions to predict the final 3D affordance map. We further integrate our affordance grounding module with large language models for motion planning, enabling enhanced spatial understanding capabilities compared to reasoning solely from images or point clouds in robotic manipulation tasks.

II. METHOD

A. Overview

Our affordance grounding pipeline consists of two components which is summarized in Fig 1. First, we construct 3D consistent semantic features from DINOv2 for object point clouds. Subsequently, our affordance grounding module takes these semantically-enriched point clouds as input and predicts the corresponding affordance maps. To enable robotic manipulation, we last leverage LLM for constraint function generation, which will be optimized during planning.

B. Semantic Point Cloud Construction

To construct our 3D feature field for point cloud scenes, we draw inspiration from the approach, D³Field, proposed by Wang et al. [9] for projecting 2D semantic features into

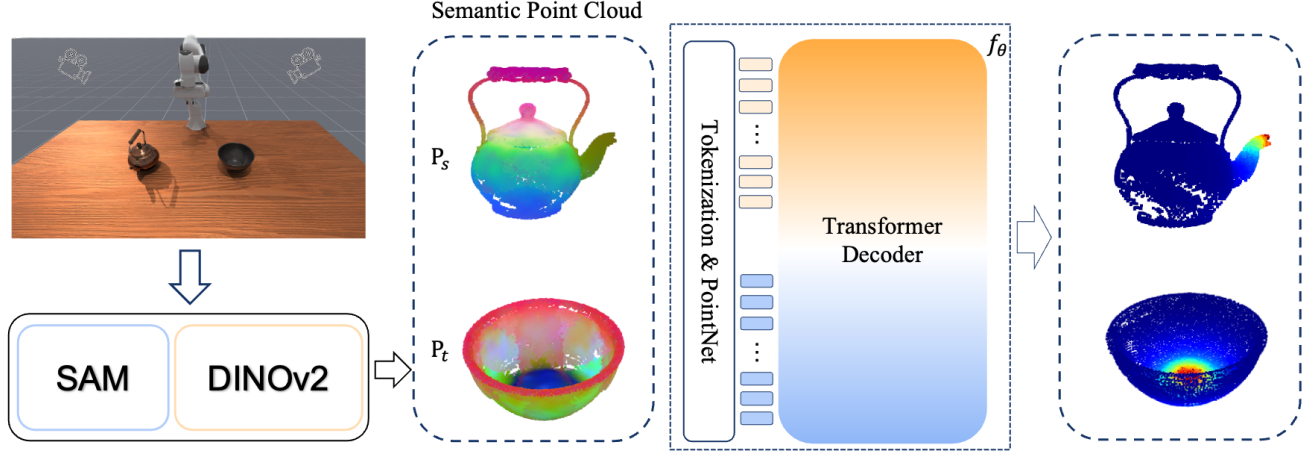


Fig. 1: **Framework pipeline.** We first using SAM [14] to get object point cloud from multi-view observations; we then constrcut semantic point cloud using DINOv2; last, we employ a bi-directional transformer decoder for fianl prediction.

3D space using DINOv2 [5]. Multi-view RGBD images are processed to extract DINOv2 features, which are then projected onto arbitrary 3D coordinates by mapping them to each camera’s image space, interpolating features, and fusing them across views. Specifically, for a 3D point $\mathbf{x}$, D³Field [9] compute its projection $\mathbf{u}_i$ in the i -th camera view, determine the truncated depth difference $d_i = r_i - r'_i$ (where r_i is sensor-captured depth and r'_i is the interpolated depth), and assign visibility v_i and weight w_i to prioritize points near the surface. These weights guide the fusion of semantic features $\mathbf{f}_i$ and instance masks $\mathbf{p}_i$ across K views, yielding a unified 3D descriptor field. We adapt this method to our point cloud representation by aligning multi-view RGB observations with the point cloud geometry, projecting DINOv2 features onto the 3D points, and fusing them to encode semantic information directly onto the point cloud structure. This approach enables efficient and generalizable feature representation without additional training, supporting robust scene understanding in our application.

C. One-Shot Affordance Grounding

To enable effective affordance prediction from point clouds, we propose a cross-object decoder to predict affordance. Our network processes paired point clouds representing a source object $\mathbf{P}_{\text{src}} \in \mathbb{R}^{B \times N \times 3}$ and a target object $\mathbf{P}_{\text{tgt}} \in \mathbb{R}^{B \times N \times 3}$, alongside corresponding visual features extracted from DINOv2

Our model f_θ comprises two primary components: (1) a **Point Cloud Encoder**, (2) a **Bi-direction transformer decoder**.

a) Point Cloud Encoder: The *PointCloudEncoder* first jointly processes concatenated point clouds and their DINOv2-derived features. Given the combined input $\mathbf{P} = [\mathbf{P}_{\text{src}}, \mathbf{P}_{\text{tgt}}] \in \mathbb{R}^{2B \times N \times 3}$ and features $\mathbf{F} = [\mathbf{F}_{\text{src}}, \mathbf{F}_{\text{tgt}}] \in \mathbb{R}^{2B \times N \times 1024}$, we first tokenize the input using farthest point sampling (fps) to select T patch centers, then employ k -nearest neighbors (kNN) for each patch center to construct the corresponding patches. To effectively aggregate local

geometric information into compact patch-level feature representations, we employ a PointNet architecture for feature extraction. This process finally produces tokenized input $\mathbf{Z} \in \mathbb{R}^{2B \times T \times 512}$.

b) Bi-directional Transformer Decoder: We apply a bidirectional transformer decoder to effectively capture geometric and semantic contextual dependencies across source and target objects. Specifically, we implement a cross-attention mechanism to enable dynamic feature interaction between object pairs. Our multi-head attention module with 8 heads captures these contextual dependencies:

$$\begin{aligned} \mathbf{A}_{\text{src}} &= \text{CrossAttention}(\mathbf{Z}_{\text{src}}, \mathbf{Z}_{\text{tgt}}, \mathbf{Z}_{\text{tgt}}), \\ \mathbf{A}_{\text{tgt}} &= \text{CrossAttention}(\mathbf{Z}_{\text{tgt}}, \mathbf{Z}_{\text{src}}, \mathbf{Z}_{\text{src}}). \end{aligned}$$

This bidirectional interaction module effectively encodes the complementary affordance relationships between interacting objects.

To generate per-point predictions, we interpolate the final patch-level embeddings back to individual points in the original point clouds using nearest-neighbor interpolation based on distance to patch centroids $\mathbf{C}$. This process yields dense point-level embeddings $\mathbf{E}_{\text{src}}, \mathbf{E}_{\text{tgt}} \in \mathbb{R}^{B \times N \times 512}$. These dense embeddings are concatenated and processed through a lightweight MLP projection head to produce the final affordance map $\mathbf{S} \in [0, 1]^{2B \times K \times N}$, representing predicted interaction probabilities, where the dimension K corresponds to separate channels for each affordance type prediction.

c) Training and Optimization: In training, we optimize our affordance grounding network using the binary cross-entropy (BCE) loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$$

During inference, the network directly outputs per-point affordance predictions, facilitating effective robotic manipulation planning and execution in real-world scenarios.

D. Affordance-Based Planning with LLMs

To leverage the predicted affordance to improve manipulation tasks, we further integrate our 3D affordance representation with large language models (LLMs) to serve as planners for low-level action execution. Specifically, we will enhance LLMs’ capability to comprehend and reason about object interactions by generating task-specific constraint functions for motion planning. Given a source object point cloud $\mathbf{P}_{\text{src}}$ and a target object point cloud $\mathbf{P}_{\text{tgt}}$, along with their respective affordance maps $\mathbf{A}_{\text{src}}$ and $\mathbf{A}_{\text{tgt}}$, we optimize a 6-DoF transformation $\mathbf{T} \in \mathbb{R}^{4 \times 4}$ that aligns objects appropriately for the intended task.

a) Constraint Generation with LLMs: We leverage the common sense reasoning capabilities of LLMs to translate object affordance and geometry representation into explicit constraints through code generation. This approach abstracts semantic affordance instructions (e.g., “pouring” or “cutting”) into concrete geometric constraints such as alignment, orientation, and spatial relations. The generated constraints include:

- **Alignment Constraint** $S_{\text{align}}(\mathbf{T})$: Ensures optimal alignment between source and target high-affordance regions.
- **Positional Constraint** $S_{\text{pos}}(\mathbf{T})$: Enforces spatial relationships based on task semantics.
- **Orientalional Constraint** $S_{\text{orient}}(\mathbf{T})$: Ensures objects maintain specific orientations (e.g., tilted or perpendicular) appropriate for the manipulation.
- **Collision Constraint** $S_{\text{coll}}(\mathbf{T})$: Prevents unrealistic penetrations or collisions between objects.
- **Stability Constraint** $S_{\text{stab}}(\mathbf{T})$: Ensures stable object placement post-manipulation.

Each constraint is guided by task-specific semantic reasoning of the LLM, making our approach robust across varied manipulation scenarios.

b) Transformation Optimization: We target the planning problem of finding the optimal transformation applied to a source object for a given task. Specifically, we formulate this as a constrained optimization over the rigid-body transformation $\mathbf{T} \in SE(3)$ applied to the source object. Our goal is to determine the optimal $\mathbf{T}$ that minimizes a composite cost function, which evaluates both the plausibility and feasibility of the resulting interaction configuration. Formally, we solve:

$$\begin{aligned} \min_{\mathbf{T} \in SE(3)} \quad & w_{\text{align}} S_{\text{align}}(\mathbf{T}) + w_{\text{pos}} S_{\text{pos}}(\mathbf{T}) + w_{\text{orient}} S_{\text{orient}}(\mathbf{T}) \\ & + w_{\text{coll}} S_{\text{coll}}(\mathbf{T}) + w_{\text{stab}} S_{\text{stab}}(\mathbf{T}) \\ \text{s.t.} \quad & \mathbf{T} \text{ satisfies task-specific constraints} \end{aligned} \quad (1)$$

This optimization can be solved using gradient-based methods (e.g., Adam [15] or L-BFGS [16]) with numerical gradients computed over the transformed geometry.

III. EXPERIMENTS

We evaluate our method for both affordance prediction and robotic manipulation tasks. We aim to answer three key research questions:

- How effectively does our method perform in object-to-object affordance grounding tasks?
- To what extent can our method generalize when trained with only a single example for each affordance type?
- How effectively can our method improve downstream LLM-based planning?

We address the first two research questions by evaluating our model’s accuracy and generalization capabilities in Sec III-A, and we investigate the third question through validation experiments in robotic manipulation scenarios presented in Sec III-B.

A. Affordance Grounding

a) Baselines: We evaluate our affordance grounding module against 2 baselines: **IAGNet** [17] and **RoboPoint** [18], comprising a 3D point cloud affordance grounding method and a VLM-based affordance method. We adopt four metrics during evaluation: **aIOU** [19], **SIMilarity** [20], **MAE** [21], and **AUC** [22].

b) Results: Tab I shows the quantitative comparison. We also present our qualitative examples in Fig 2. Each row in Fig 2 represents a distinct affordance type, with three different affordance types illustrated. For each affordance type, we present two corresponding rows: the upper showing the source object and the lower displaying the target object. We observed that RoboPoint, as a vision-language model, demonstrates capability in object localization but lacks the precision to infer fine-grained affordance regions on objects. IAGNet exhibits the strongest performance among all baselines but suffers significantly from overfitting due to the one-shot training paradigm. Our method significantly outperforms all baselines and demonstrates robust generalization capabilities.

Method	↑ IOU	↑ SIM	↓ MAE	↑ AUC
IAGNet	14.81	0.5574	0.1402	73.30
RoboPoint	11.84	0.4376	0.3344	59.78
Ours	17.82	0.6387	0.0612	96.00

TABLE I: **Quantitative comparison on object-to-object affordance grounding.**

B. Affordance-based Manipulation

a) Baselines: We evaluate LLM-based planning against an ablated version of our method that performs planning directly from object point clouds. Tab II compares the success rate for each task. We evaluate our approach against the baseline across six manipulation tasks, conducting ten trials per task and recording success rate as the primary evaluation metric.

b) Results: According to quantitative results, affordance, as a mid-level representation, significantly enhances manipulation success rates. In common tasks requiring two-object interaction, our method achieves approximately 80% success rate, while the baseline exhibits considerably poorer performance due to its inability to recognize functional properties from object point clouds. In more complex and extended-horizon tasks such as hanging and plugging, the

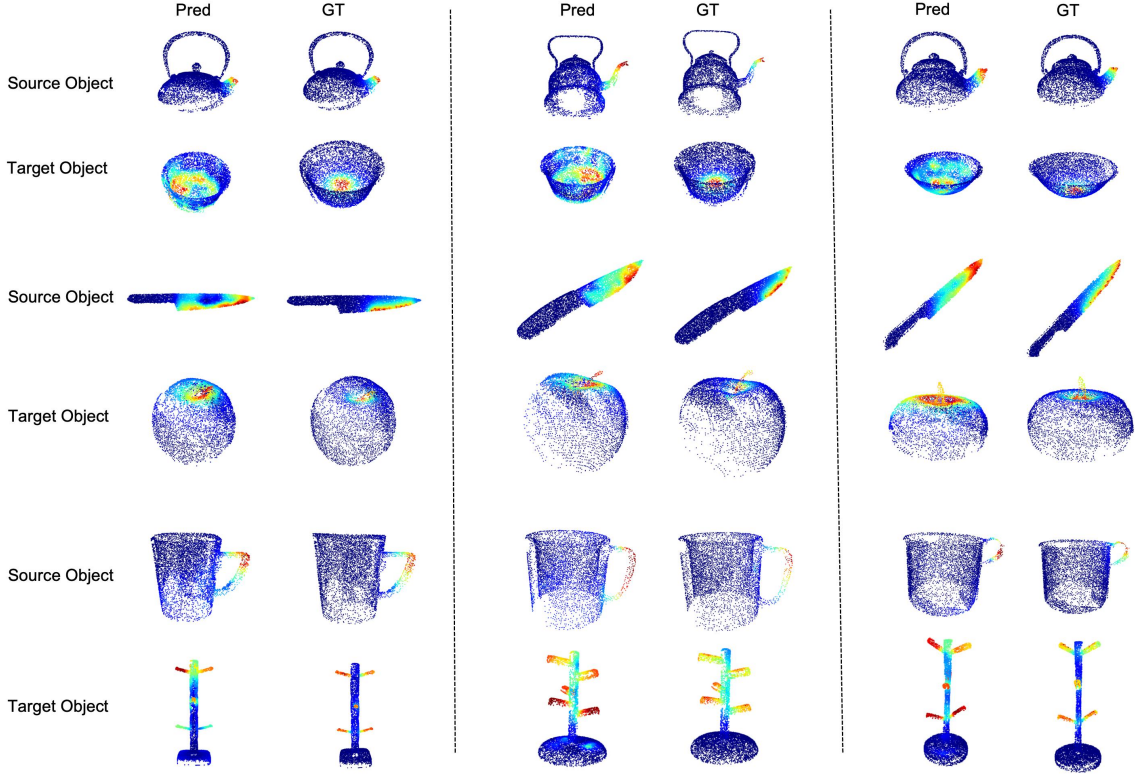


Fig. 2: **Qualitative examples of object-to-object affordance prediction.** Each row corresponds to a specific affordance type, with each affordance evaluated across three different test object pairs (columns).

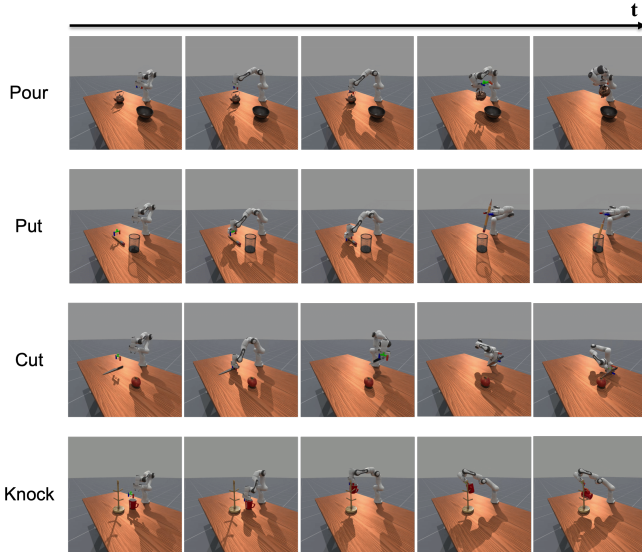


Fig. 3: **Example Manipulation Results using O³Afford.** Each row corresponds to a manipulation task, with the x-axis representing time to illustrate the progression of the task.

baseline fails in all trials, whereas our method maintains approximately 50% success rate. We present qualitative results in Fig 3. Each row depicts a distinct manipulation task, while each column illustrates the sequential progression of the task execution. As illustrated in Fig 3, our tasks involve extended time horizons and require precise contact

for successful completion. Our results demonstrate that affordance representation significantly enhances LLMs’ spatial reasoning capabilities, as evidenced by the resulting coherent and purposeful manipulation sequences.

Method	Pouring	Hanging	Pressing	Putting	Cutting
Baseline	2/10	0/10	5/10	5/10	4/5
Ours (%)	8/10	5/10	9/10	8/10	9/10

TABLE II: **Success Rate of Manipulation Tasks.** Each task is evaluated over 10 trials.

IV. CONCLUSION

We presented O³Afford, a novel one-shot object-to-object affordance grounding framework tailored for robotic manipulation. Our approach distills rich part-aware semantic features from vision foundation models onto 3D point clouds, enabling effective intra-class generalization capability with minimal supervision. By leveraging bi-directional context between source and target objects, our method captures more complex interaction patterns, such as pouring and cutting, which are essential yet underexplored in prior work. Furthermore, the integration with LLMs facilitates spatial reasoning beyond traditional visual or geometric cues alone. Experimental results demonstrate that O³Afford achieves strong intra-class generalization capability, making it a promising step toward scalable, data-efficient affordance learning in real-world robotic systems.

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